	Academic Semester	Fall 2026-2027	**Academic level**	3
	Course Code	MT301	**Exam**	Quiz 1 — Model B
	Course Title	Probability and Statistics I		
	Time Allowed	30 Minutes	**Total Points**	10
	Course Instructors	Dr. Mai Farag , Dr. Walid Abd-Elraheem		

Student Name: _____ **Student ID:** _____

*Answer **all** questions. Calculators are allowed. Show your work in Questions 2 and 3; give probabilities as simplified fractions.*

Question 1 — Choose the correct answer (4 points)

One point each: ½ for the correct letter, ½ for a correct reason. Circle the letter and write why you chose it in the box.

1. A student has 5 shirts, 3 pairs of trousers and 4 pairs of shoes. How many different outfits of one shirt, one pair of trousers and one pair of shoes are possible?

A) 12 **B)** 60 **C)** 20 **D)** 45

Why? (reason or calculation)

2. Eight runners take part in a race. In how many ways can the gold, silver and bronze medals be awarded?

A) 336 **B)** 56 **C)** 512 **D)** 24

Why? (reason or calculation)

3. Two fair dice are rolled. The probability that the sum of the two faces is 9 is

A) $\frac{1}{6}$ **B)** $\frac{1}{12}$ **C)** $\frac{5}{36}$ **D)** $\frac{1}{9}$

Why? (reason or calculation)

4. A fair coin is tossed three times. The probability of getting **at least** two heads is

A) $\frac{3}{8}$ **B)** $\frac{1}{4}$ **C)** $\frac{1}{2}$ **D)** $\frac{7}{8}$

Why? (reason or calculation)

Question 2 — Counting (3 points)

A committee of 3 people is to be chosen from 7 men and 4 women.

(a) In how many ways can the committee be chosen? [1.5 pt]

(b) In how many ways can it be chosen if it must contain **exactly 1 woman**? [1.5 pt]

Question 3 — Probability (3 points)

A 3-digit number is formed at random from the digits $\{1, 2, 3, 4, 5\}$, with **no digit repeated**.

(a) Find the probability that the number is **even**. [1.5 pt]

(b) Find the probability that the number is **greater than 300**. [1.5 pt]

— End of quiz · Good luck —